# Vansh Pratap Singh Ranawat

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LinkedIn | GitHub | LeetCode | Portfolio

## Professional Summary

Computer Science and Engineering graduate from MIT ADT University with hands-on experience building full-stack web applications using React.js, JavaScript, Node.js, Express.js, and MongoDB. Proficient in REST API development, JWT authentication, database design, and Git-based version control. Strong foundation in Java, Object-Oriented Programming, Data Structures and Algorithms, and software engineering fundamentals. Experienced in developing, testing, and optimizing applications, with a focus on clean, maintainable code and reliable software solutions. Seeking a Software Developer or Full-Stack Developer role to contribute to real-world projects and build scalable applications.

## Technical Skills

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| **Frontend:** | React.js, JavaScript (ES6+), HTML5, CSS3, Responsive Design, Component-Based Architecture, State Management |
| **Backend & APIs:** | Java, Node.js, Express.js, REST API Design, JWT Authentication |
| **Databases:** | MongoDB, SQL |
| **Testing & Tools:** | Postman (API testing), Git, GitHub, GitHub Actions (CI/CD), VS Code |
| **Fundamentals:** | Object-Oriented Programming, Data Structures & Algorithms, Debugging, SDLC |

## Selected Projects

**Lost and Found Web Application** Mar 2025 – May 2025
*React, Node.js, Express, MongoDB, REST APIs, GitHub Actions* Live Demo | GitHub

- Delivered a full-stack lost-and-found platform where users report and manage items through CRUD workflows, with a React frontend and a Node.js/Express REST backend.
- Created 15+ reusable React components with centralized state management to reduce duplicated UI logic and keep the codebase maintainable.
- Implemented 12+ REST API endpoints in Express using an MVC structure, consumed by the React frontend.
- Secured 8+ private API routes with JWT authentication and custom authorization middleware.
- Designed the MongoDB schema and indexed 4 frequently queried fields, reducing average API response time by 30%.
- Wrote 25+ Postman test cases covering API behavior and data integrity, catching regressions before deployment.
- Set up a GitHub Actions CI/CD pipeline to automate build, test, and deployment.

**Receipt OCR Data Extraction System** Jan 2026 – Feb 2026
*Python, OpenCV, Regex, Linux, Git* GitHub

- Engineered a modular Python pipeline that extracts store name, date, and total amount from receipt images into structured JSON.
- Applied OpenCV preprocessing (Gaussian blur, Otsu thresholding), improving character recognition accuracy by over 30% on noisy scans.
- Wrote regex-based validation that normalizes 10+ date formats and distinguishes subtotal, tax, and total amounts.
- Added confidence scoring and edge-case handling, reaching 92% extraction reliability.
- Structured the code into 4 reusable modules with a command-line interface supporting batch processing of up to 100 images.

**Tamper-Proof Certificate Verification System** Sep 2025 – Nov 2025
*Solidity, Web3.js, MetaMask, REST APIs* Live Demo | GitHub

- Designed a blockchain-based certificate verification system using 3 Solidity smart contracts to store tamper-evident records on-chain; SHA-256 document fingerprints cut verification time from minutes to under 5 seconds.
- Integrated Web3.js and MetaMask for wallet-based authentication and transaction signing.
- Created REST APIs to issue and validate credentials, tested against 200+ sample records; removed redundant on-chain writes, lowering average gas use by 18%.

## Education

**Bachelor of Technology, Computer Science and Engineering** 2022 – 2026
MIT ADT University, Pune

**Class XII (CBSE), Science (PCM + Computer Science)** 2020 – 2021
St. Anselm Sr. Sec. School, Bhilwara

## Certifications

**Foundations of Prompt Engineering**, AWS Training & Certification July 2026 | View Certificate

**Introduction to Software Engineering**, IBM (Coursera) Nov 2024 | View Certificate